Curriculum Vitae — Vijayakumar



Name	Vijayakumar Muni (Official name: Vijayakumar)
Designation	Assistant Professor
Affiliation	Department of Mathematics, Smt. Indira Gandhi Government First Grade Women's College Sagara, Sagara (PIN: 577401), Karnataka, India
Contact No.	+91 81054 89930
E-mail	`vijayakumarmuni[zero][at]gmail[dot]com`
Born	1989-12-10
Place of birth	Gurumitkal, Karnataka, India
Human languages	English, Hindi, Kannada
Software & Tools	LaTeX; MATLAB; MATHEMATICA (learning); MS Office: Word, Excel, PowerPoint; Python (learning); R (learning)

Education

2015-07-27/2019-07-05	PhD	Department of Mathematics, Indian Institute of Space Science and Technology, Valiamala (PIN: 695547), Kerala, India, under the supervision of Dr. Raju K George for the thesis titled *"Studies on Controllability of Dynamical Systems with Impulses and Time-Delay Controls"*
2010-07/2012-07-27	MSc (with 84.8%)	Mathematics, Gulbarga University, Kalaburagi (PIN: 585106), Karnataka, India
2007-07/2010-07-25	BSc (with 85.24%)	Mathematics, Physics, and Chemistry, Sharnbasveshwar College of Science (affiliated to Gulbarga University), Kalaburagi (PIN: 585103), Karnataka, India

Academic Honors

1. (offline) Fellow, "Summer Research Fellowship Program-2025", supervised by Kiran Kumari (Department of Electrical Engineering), Indian Institute of Science, Bengaluru (PIN: 560012), Karnataka, India, 2025-09/2025-10, IASc–INSA–NASI, India
2. (online) Fellow, "Summer Faculty Research Fellow Programme under Continuing Education Programme-2020", supervised by Konijeti Sreenadh (Department of Mathematics), 2020-06-08/2020-08-08, Indian Institute of Technology Delhi, New Delhi (PIN: 110016), NCT of Delhi, India
3. "CSIR–UGC–NET for Lectureship in Mathematical Sciences-Jun 2015", AIR-36 (result declared on 2015-09-30; certificate received on 2016-01-27)
4. "CSIR–UGC–NET for Lectureship in Mathematical Sciences-Dec 2014", AIR-26 (result declared on 2015-03-27; certificate received on 2015-06-25)
5. "Graduate Aptitude Test in Engineering in Mathematics-2015", AIR-716, GATE score-365 (result declared on 2015-03-27)
6. "Karnataka State Eligibility Test for Lectureship in Mathematical Science-2013" (result declared on 2014-04-28)
7. "Prof. V. P. Rathod Gold Medal" for securing highest marks in Numerical Analysis in MSc Programme, 2013-03-06
8. "6th Rank in Bachelor of Science", Gulbarga University, 2011-01

Research Interests

1. Multi-agent networked systems
2. Control-theoretic properties
3. Impulsive systems
4. Time-delay systems

Online Profiles

Academic

1. (Scopus) `https://www.scopus.com/authid/detail.uri?authorId=57204153141`
2. (Open Researcher and Contributor ID) `https://orcid.org/0000-0001-5749-8119`

3. (Google Scholar) `https://scholar.google.com/citations?user=RnnY47kAAAAJ&hl=en`
4. (ResearchGate) `https://www.researchgate.net/profile/Vijayakumar_Muni`
5. (LinkedIn) `https://www.linkedin.com/in/vijayakumar-muni-727b1937/`
6. (Mathematics Genealogy Project) `https://www.genealogy.math.ndsu.nodak.edu/id.php?id=260539`
   - Hilbert number: 5
   - Erdős number: 6
   - Lipschitz number: 8
   - Dirichlet number: 9
   - Fourier number: 10
   - Gauss number: 10
   - Poisson number: 10
   - Lagrange number: 11
   - Laplace number: 11
   - d'Alembert number: 12
   - Euler number: 12
7. AMS Mathematical Reviews collaborative distance to:
   - M. Atiyah: 5
   - P. Erdős: 5
   - J. Milnor: 5
   - I. Singer: 5
   - T. Tao: 5
   - E. Witten: 5
   - M. Bhargava: 6
   - A. Einstein: 6
   - S. Hawking: 6
   - W. Heisenberg: 6
   - J. Nash: 6
   - R. Penrose: 6
   - E. Schrödinger: 6
   - J. P. Serre: 6
   - A. Wiles: 6
   - D. Hilbert: 7
   - C. F. Gauss: 8
   - S. Ramanujan: 8

Social Media

1. (YouTube) `https://www.youtube.com/@vijayakumarmuni5887`
2. (Facebook) `https://www.facebook.com/vijay.muni.10`
3. (Instagram) `https://www.instagram.com/vijayakumarmuni/`
4. (X) `https://x.com/munivijay11`

Publications

Journal Articles

1. B. Aadhithyan, Vijayakumar S. Muni*, V. S. Athira, Muhammed Rafeek Kallu Vetty, and Gudala Janardhana Reddy, *"Sensor–non-sensor observability of directed and weighted signed networks under heterogeneous dynamics in n-dimensional space"*, International Journal of Control (Taylor & Francis, 5 Howick Place, London (postcode: SW1P 1WG), United Kingdom), vol. 99, no. 8, pp. 2616–2637, 2026-06-01, DOI: 10.1080/00207179.2026.2679225, P-ISSN: 0020-7179, E-ISSN: 1366-5820, Indexed in: Science Citation Index Expanded™ (Clarivate™), Scopus® (Elsevier), MathSciNet® (AMS), zbMATH; Impact Factor: 1.6 (JCR-2024), CiteScore: 4.5 (Scopus®-2024); Q3|Automation & Control Systems (JCR-2024)
2. Athira V. S., Vijayakumar S. Muni*, Kallu Vetty Muhammed Rafeek, Aadhithyan B., and Gudala Janardhana Reddy, *"OBSERVABILITY STUDY IN WEIGHTED AND DIRECTED SIGNED MULTI-AGENT NETWORKS OF SINGLE- AND DOUBLE-INTEGRATOR DYNAMICAL AGENTS"*, Indian Journal of Mathematics (Allahabad Mathematical Society, 10, C. S. P. Singh Marg, Prayagraj (PIN: 211001), Uttar Pradesh, India), vol. 67, no. 2, pp. 169–201, 2025-08, P-ISSN: 0019-5324, Indexed in: Scopus® (Elsevier), MathSciNet® (AMS), zbMATH; Impact Factor: NA (JCR-2024), CiteScore: 0.4 (Scopus®-2024); Q4|Mathematics (miscellaneous) (Scimago Journal & Country Rank-2024)

3. Vijayakumar S. Muni*, Kallu Vetty Muhammed Rafeek, Gudala Janardhana Reddy, and Deena Sunil Sharanappa, *"Signed networks with Laplacian feedback: Observability and minimal sensor problem"*, Bulletin of the Polish Academy of Sciences Technical Sciences (Division IV Engineering Sciences of the Polish Academy of Sciences, Polska Akademia Nauk, pl. Defilad 1, Warszawa (postal code: 00-901), Poland), vol. 73, no. 4, pp. e154061, 2025-07-02 (published online on 2025-04-01), DOI: 10.24425/bpasts.2025.154061, P-ISSN: 0239-7528, E-ISSN: 2300-1917, Indexed in: Science Citation Index Expanded™ (Clarivate™), Scopus® (Elsevier); Impact Factor: 1.2 (JCR-2024), CiteScore: 2.7 (Scopus®-2024); Q3|Multidisciplinary Engineering (JCR-2024)

4. Vijayakumar S. Muni*, Kallu Vetty Muhammed Rafeek, Gudala Janardhana Reddy, and Raju K. George, *"OBSERVABILITY OF MULTI-AGENT NETWORKS OVER RANDOM-WALK NORMALISED LAPLACIAN DYNAMICS"*, Indian Journal of Mathematics (Allahabad Mathematical Society, 10, C. S. P. Singh Marg, Prayagraj (PIN: 211001), Uttar Pradesh, India), vol. 65, no. 3, pp. 295–322, 2023-12, P-ISSN: 0019-5324, Indexed in: Scopus® (Elsevier), MathSciNet® (AMS), zbMATH; Impact Factor: NA (JCR-2024), CiteScore: 0.4 (Scopus®-2024); Q4|Mathematics (miscellaneous) (Scimago Journal & Country Rank-2024)

5. Vijayakumar S. Muni*, Kallu Vetty Muhammed Rafeek, Athira V.S., and Gudala Janardhana Reddy, *"Controllability of consensus of multi-agent networks over heterogeneous dynamics"*, Results in Control and Optimization (Elsevier B.V., Radarweg 29, Amsterdam (postal code: 1043 NX), Netherlands), vol. 12, pp. 100272, 2023-09 (published online on 2023-07-19), DOI: 10.1016/j.rico.2023.100272, E-ISSN: 2666-7207, Indexed in: Emerging Sources Citation Index™ (Clarivate™), Scopus® (Elsevier); Impact Factor: NA (JCR-2024), CiteScore: 5.1 (Scopus®-2024); Q2|Control and Optimization (Scimago Journal & Country Rank-2024)

6. Athira V. S., Vijayakumar S. Muni*, Kallu Vetty Muhammed Rafeek, and Gudala Janardhana Reddy*, *"Controllability of consensus heterogeneous multi-agent networks over continuous time scale"*, Control and Cybernetics (Systems Research Institute of the Polish Academy of Sciences, Newelska 6, Warszawa (postal code: 01-447), Poland), vol. 52, no. 2, pp. 199–245, 2023-06 (published online on 2024-04-11), DOI: 10.2478/candc-2023-0037, P-ISSN: 0324-8569, E-ISSN: 2720-4278, Indexed in: Scopus® (Elsevier), MathSciNet® (AMS), zbMATH; Impact Factor: NA (JCR-2024), CiteScore: 0.9 (Scopus®-2024); Q4|Control and Systems Engineering (Scimago Journal & Country Rank-2024)

7. Vijayakumar S. Muni*, Kallu Vetty Muhammed Rafeek, Gudala Janardhana Reddy, and Raju K. George, *"On the Selection of Leaders for the Controllability of Multi-agent Networks"*, Bulletin of the Iranian Mathematical Society (Springer Nature on behalf of the Iranian Mathematical Society, Gewerbestraße 11, Cham (postcode: 6330), Switzerland), vol. 48, no. 6, pp. 3141–3183, 2022-12-01 (published online on 2022-02-24), DOI: 10.1007/s41980-022-00683-2, P-ISSN: 1017-060X, E-ISSN: 1735-8515, Indexed in: Science Citation Index Expanded™ (Clarivate™), Scopus® (Elsevier), MathSciNet® (AMS), zbMATH; Impact Factor: 0.7 (JCR-2024), CiteScore: 1.9 (Scopus®-2024); Q2|Mathematics (JCR-2024)

8. Vijayakumar S. Muni* and Raju K. George, *"On Controllability of Networked Higher Dimensional Impulsive Systems"*, Bulletin of the Iranian Mathematical Society (Springer Nature on behalf of the Iranian Mathematical Society, Gewerbestraße 11, Cham (postcode: 6330), Switzerland), vol. 47, no. 6, pp. 1947–1968, 2021-12-01 (published online on 2021-01-29), DOI: 10.1007/s41980-020-00481-8, P-ISSN: 1017-060X, E-ISSN: 1735-8515, Indexed in: Science Citation Index Expanded™ (Clarivate™), Scopus® (Elsevier), MathSciNet® (AMS),

zbMATH; Impact Factor: 0.7 (JCR-2024), CiteScore: 1.9 (Scopus®-2024); Q2|Mathematics (JCR-2024)

9. Vijayakumar S. Muni* and Raju K. George, *"CONTROLLABILITY OF LINEAR IMPULSIVE SYSTEMS—AN EIGENVALUE APPROACH"*, Kybernetika (Nakladatelství Academia Centre of Administration and Operations of the Czech Academy of Sciences & Institute of Information Theory and Automation of the Czech Academy of Sciences, Pod Vodárenskou věží 4, Prague (postal code: 182 08), Czech Republic), vol. 56, no. 4, pp. 727–752, 2020-10-11, DOI: 10.14736/kyb-2020-4-0727, P-ISSN: 0023-5954, E-ISSN: 1805-949X, Indexed in: Science Citation Index Expanded™ (Clarivate™), Scopus (Elsevier®), MathSciNet® (AMS), zbMATH; Impact Factor: 0.9 (JCR-2024), CiteScore: 2.1 (Scopus®-2024); Q4|Cybernetics Computer Science (JCR-2024)

10. Vijayakumar S. Muni* and Raju K. George, *"Controllability of semilinear impulsive control systems with multiple time delays in control"*, IMA Journal of Mathematical Control & Information (Oxford University Press on behalf of Institute of Mathematics & its Applications, Great Clarendon Street, Oxford (postcode: OX2 6DP), United Kingdom), vol. 36, no. 3, pp. 869–899, 2019-09-16 (published online on 2018-03-26), DOI: 10.1093/imamci/dny011, P-ISSN: 0265-0754, E-ISSN: 1471-6887, Indexed in: Science Citation Index Expanded™ (Clarivate™), Scopus (Elsevier®), MathSciNet® (AMS), zbMATH; Impact Factor: 1.6 (JCR-2024), CiteScore: 3 (Scopus®-2024); Q2|Applied Mathematics (JCR-2024)

11. Vijayakumar S. Muni* and Raju K. George, *"CONTROLLABILITY OF LINEAR IMPULSIVE MATRIX LYAPUNOV DIFFERENTIAL SYSTEMS WITH DELAYS IN THE CONTROL FUNCTION"*, Kybernetika (Nakladatelství Academia Centre of Administration and Operations of the Czech Academy of Sciences & Institute of Information Theory and Automation of the Czech Academy of Sciences, Pod Vodárenskou věží 4, Prague (postal code: 182 08), Czech Republic), vol. 54, no. 4, pp. 664–698, 2018-09-24, DOI: 10.14736/kyb-2018-4-0664, P-ISSN: 0023-5954, E-ISSN: 1805-949X, Indexed in: Science Citation Index Expanded™ (Clarivate™), Scopus® (Elsevier), MathSciNet® (AMS), zbMATH; Impact Factor: 0.9 (JCR-2024), CiteScore: 2.1 (Scopus®-2024); Q4|Cybernetics Computer Science (JCR-2024)

12. Vijayakumar S. Muni*, Venkatesan Govindaraj, and Raju K. George, *"CONTROLLABILITY OF FRACTIONAL ORDER SEMILINEAR SYSTEMS WITH A DELAY IN CONTROL"*, Indian Journal of Mathematics (Allahabad Mathematical Society, 10, C. S. P. Singh Marg, Prayagraj (PIN: 211001), Uttar Pradesh, India), vol. 60, no. 2, pp. 311–335, 2018-08, P-ISSN: 0019-5324, Indexed in: Scopus® (Elsevier), MathSciNet® (AMS), zbMATH; Impact Factor: NA (JCR-2024), CiteScore: 0.4 (Scopus®-2024); Q4|Mathematics (miscellaneous) (Scimago Journal & Country Rank-2024)

Scholarly and Educational Resources

1. Vijayakumar Muni, Althaf M., Kallu Vetty Muhammed Rafeek, and Venkatesan Govindaraj, *"A Mathematical Framework for Controllability of Single-Agent Systems Modelled as a Point-Particle Governed by Linear Dynamics in Continuous-Time (Technical Report: A Pedagogical Exposition)"*, Zenodo (CERN (postal code: 1211), Geneva 23, Switzerland), 2026-05-29, DOI: 10.5281/zenodo.20487021, Indexed in: OpenAIRE

2. Vijayakumar Muni, *"Disparity to Consensus: A Social Network as an Example"*, Zenodo (CERN (postal code: 1211), Geneva 23, Switzerland), 2025-12-25, DOI: 10.5281/zenodo.18049144, Indexed in: OpenAIRE

3. Vijayakumar Muni and Kiran Kumari, *"Fundamental Notions in a Multi-Agent Network (each agent as a spatial point when it is physical) under Ordinary Differential Dynamics (Technical Report: A Pedagogical Exposition)"*, Zenodo (CERN (postal code: 1211), Geneva 23, Switzerland), 2025-10-23, DOI: 10.5281/zenodo.17438770, Indexed in: OpenAIRE

4. Vijayakumar Muni, *"***E***-Tracking Controllability (Special Case: Trajectory Controllability): Part-II"*, Zenodo (CERN (postal code: 1211), Geneva 23, Switzerland), 2025-07-05, DOI: 10.5281/zenodo.15811114, Indexed in: OpenAIRE

5. Vijayakumar Muni, *"Synchronization of Self-Driving Cars: Velocity Consensus"*, Zenodo (CERN (postal code: 1211), Geneva 23, Switzerland), 2025-02-24, DOI: 10.5281/zenodo.15255747, Indexed in: OpenAIRE

6. Vijayakumar Muni, *"***E***-Tracking Controllability (Special Case: Trajectory Controllability): Part-I"*, Zenodo (CERN (postal code: 1211), Geneva 23, Switzerland), 2025-02-24, DOI: 10.5281/zenodo.15252115, Indexed in: OpenAIRE

7. Developer of YouTube videos on

   - "Every group of order less than or equal to five is abelian", 2020-12-31, URL: https://youtu.be/W0eRwfX8Le4
   - "Fermat's Last Theorem—the 300-year-old mystery", 2020-05-06, URL: https://youtu.be/0eoVYyDbhdY
   - "Trigonometric functions and their applications", 2020-05-06, URL: https://youtu.be/AARBonQsd00

Media Coverage

1. Deena Sunil and Vijayakumar Muni, "साधारण कंप्यूटर जिस समस्या को वर्षों में हल करेगा उसे क्वांटम कंप्यूटर मिनटों में ही सुलझा सकता है", Dainik Bhaskar, Jabalpur, Madhya Pradesh, India, 2026-02-28

2. "रानी दुर्गावती विश्वविद्यालय में अंतरराष्ट्रीय कांफ्रेंसका आरंभ", Aaj ka Bharat, Jabalpur, Madhya Pradesh, India, 2026-02-26, URL: https://youtu.be/PssFCA9XH7U?t=389

Invited Academic Talks

1. (offline) *"Controllability: Single-agent to multi-agent systems" and "Data envelopment analysis: Efficiency through optimization"*, 2026-08-21/2026-08-22, Department of Mathematics, Central University of Karnataka, Kadaganchi (PIN: 585367), Karnataka, India

2. (offline) *"A mathematical framework for MAS controllability"*, International Conference on Science for a Sustainable Future: Innovation at the Interface of Pure and Applied Research, 2026-02-26/2026-02-28, Rani Durgavati University, Jabalpur (PIN: 482001), Madhya Pradesh, India

3. (online) *"Importance of Mathematics in Computer Science, AI, and Machine Learning"*, Expert Talk Series 4–Talk 2, 2026-01-10, Department of Computer Science and Engineering (AI&ML) in association with Centre for Innovation, Incubation, Research & Entrepreneurship, Kakatiya Institute of Technology & Science, Warangal (PIN: 506015), Telangana, India

4. (offline) *"Disparity to consensus: Two illustrating examples"*, International Conference on Recent Advances in Mathematical Sciences and Applications, 2025-12-19/2025-12-22, Gayatri Vidya Parishad College of Engineering (Prof. V. Lakshmikantham Institute for Advanced Studies), Visakhapatnam (PIN: 530048), Andhra Pradesh, India

5. (offline) *"Limitations of CCR & the birth of BCC"* and *"SBM model & a case study"*, One Week Capacity Building Program on Contemporary Mathematics for Data-Driven Applications, 2025-11-12/2025-11-13, Human Resource Development Centre in association with the Department of Mathematics, Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Avadi (PIN: 600055), Tamil Nadu, India

6. (offline) *"Efficiency measurement in decision making units operating under variable return to scale"* and *"Non-radial model: Slack-based measure of efficiency"*, Two-Day National Workshop on Efficiency Measurement in Data Envelopment Analysis (Hybrid), 2025-11-10/2025-11-11, Department of Mathematics, School of Advanced Studies, Vellore Institute of Technology Chennai, Chennai (PIN: 600127), Tamil Nadu, India

7. (offline) *"Python programming for DEA and Hybrid DEA-AI frameworks"*, International Workshop on Augmenting Data Envelopment Analysis with AI/ML @ South Asian University, 2025-08-03, jointly organized by Faculty of Economics, South Asian University, New Delhi (PIN: 110068), NCT of Delhi, India and Queen's Business School, Queen's University Belfast, Belfast (postcode: BT9 5EE), United Kingdom

8. (online) *"Addressing variable returns to scale in data envelopment analysis—BCC model"*, Short-Term Training Program on Frontiers in Applied Mathematics: Modelling, Computation, and Quantum-Driven Techniques, 2025-07-12, Department of Mathematics, National Institute of Technology Raipur, Raipur (PIN: 492010), Chhattisgarh, India

9. (offline) *"***E***-Tracking controllability in MAS"*, International Conference on Mathematical Analysis & its Applications—Hybrid Mode, 2025-06-28/2025-06-30, Department of Mathematics, Karnatak University, Dharwad (PIN: 580003), Karnataka, India

10. (offline) *"Agreement in social networks: A mathematical touch"*, National Seminar on Applications of Pure and Applied Mathematics, 2024-06-20, Department of Mathematics, Government First Grade College Bhalki, Bhalki (PIN: 585328), Karnataka, India

11. (offline) *"Decoding signed networks: Key observability insights"*, National Seminar on Advances in Mathematical Sciences, 2024-06-19, Department of Mathematics, Guru Nanak First Grade College, Bidar (PIN: 585403), Karnataka, India

12. (offline) *"Topological data analysis"*, National Seminar on Quantum Computing & Data Analytics (Hybrid), 2024-06-18, Department of Mathematics, Karnatak Arts, Science & Commerce College, Bidar (PIN: 585401), Karnataka, India

13. (offline) *"Dancing with dynamics: Ordinary differential equations in everyday scenarios"*, Workshop on Research Methodology (On the Eve of National Mathematics Day), 2023-12-29, Department of Mathematics, Karnatak Arts, Science & Commerce College, Bidar (PIN: 585401), Karnataka, India

14. (offline) *"Lecture on linear algebra"*, Mathematics Workshop for Undergraduate Students, 2023-10-01, Department of Mathematics, National Institute of Technology Puducherry, Karaikal (PIN: 609609), UT of Puducherry, India

15. (offline) *"Lectures on control theory"*, 2020-01-08, Department of Mathematics, Central University of Karnataka, Kadaganchi (PIN: 585367), Karnataka, India

16. (offline) *"Controllability of a LTI-complex networked systems with impulses"*, International Conference on Recent Advances in Algebra, Analysis, and Applications, 2019-12-20/2019-12-22, Department of Mathematics & Statistics, Mohanlal Sukhadia University, Udaipur (PIN: 313001), Rajasthan, India

17. (offline) *"Impulsive and delay control systems: History, mathematical achievements, and perspectives"*, International Conference on History and Recent Developments in Mathematics with Applications in Science and Technology @ Madhuben & Bhanubhai Patel Institute of Technology, 2019-12-17/2019-12-19, jointly organized by Madhuben & Bhanubhai Patel Institute of Technology, New Vallabh Vidyanagar (PIN: 388121), Gujarat, India and Indian Society for History of Mathematics, University of Delhi, New Delhi (PIN: 110007), NCT of Delhi, India

Session Chair

1. (online) "International Conference on Fractional Calculus: Theory, Applications and Numerics", 2023-01-28, Department of Mathematics, National Institute of Technology Puducherry, Karaikal (PIN: 609609), UT of Puducherry, India

2. (offline) "International Conference on Multidisciplinary Research", 2022-11-22/2022-11-24, Sharnbasva University, Kalaburagi (PIN: 585103), Karnataka, India

3. (offline) "International Conference on Recent Advances in Algebra, Analysis, and Applications", 2019-12-20/2019-12-22, Department of Mathematics & Statistics, Mohanlal Sukhadia University, Udaipur (PIN: 313001), Rajasthan, India

Academic Positions

2023-08-17/Present	Assistant Professor (on regular)	Smt. Indira Gandhi Government First Grade Women's College Sagara, Sagara (PIN: 577401), Karnataka, India
2017-08-22/2023-08-16	Assistant Professor (on regular)	Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India

2015-02-16/2015-05-29, 2014-08-18/2015-02-14	Faculty (on contract)	Central University of Karnataka, Kadaganchi (PIN: 585367), Karnataka, India

Teaching Experience

1. @ Smt. Indira Gandhi Government First Grade Women's College Sagara
   - (74604) Ordinary Differential Equations: 2024-11/2025-03, 2023-12/2024-04
   - (74614) Partial Differential Equations: 2025-04/2025-07, 2024-06/2024-09
   - (74622) Topology-I: 2025-09/2025-12, 2024-11/2025-03, 2023-12/2024-04
   - (74623) Numerical Analysis-I: 2025-09/2025-12, 2024-11/2025-03, 2023-12/2024-04
   - (74632) Topology-II: 2025-04/2025-07, 2024-06/2024-09
   - (74633) Numerical Analysis-II: 2025-04/2025-07, 2024-06/2024-09
2. @ Sri H. D. Devegowda Government First Grade College Paduvalahippe
   - (18101 HC) Algebra-I: 2021-02/2021-04, 2019-08/2019-11, 2018-08/2018-11, 2017-08/2017-11
   - (18105 SC) Linear Algebra: 2022-02-02/2022-05-31, 2021-02/2021-04, 2019-08/2019-11, 2018-08/2018-11, 2017-08/2017-11
   - (18107 SC) Graph Theory: 2021-10/2022-02
   - (18123 HC) Complex Analysis-II: 2018-01/2018-04
   - (18124 SC) Ordinary and Partial Differential Equations: 2021-05/2021-08, 2020-01/2020-08, 2019-01/2019-04, 2018-01/2018-04
   - (18141 HC) Elements of Functional Analysis: 2021-10/2022-02
   - (18142 HC) Topology-I: 2020-09/2021-02, 2019-08/2019-11, 2018-08/2018-11, 2017-08/2017-11
   - (18162 HC) Topology-II: 2021-04/2021-07, 2020-01/2020-08, 2019-01/2019-04, 2018-01/2018-04
   - (18164 SC) Advanced Graph Theory: 2021-04/2021-07, 2020-01/2020-08, 2019-01/2019-04
3. @ Central University of Karnataka
   - (MATH-202) Complex Analysis: Spring 2015
   - (MATH-402) Lebesgue Measure Theory: Spring 2015
   - (MATH-406) Integral Equations: Spring 2015
   - (MATH-103) Discrete Mathematics: Autumn 2014
   - (MATH-304) Functional Analysis: Autumn 2014

Student Supervision

1. (offline) MSc dissertation titled *"Distributed control and coordination of multi-agent systems"*, Department of Mathematics, Smt. Indira Gandhi Government First Grade Women's College Sagara, Sagara (PIN: 577401), Karnataka, India, 2025-09-01/2026-05-31
2. (online, jointly with V. Govindaraj) MSc dissertation titled *"A mathematical framework for controllability of single-agent systems modelled as a point-particle governed by linear dynamics in continuous-time"*, School of Mathematics and Statistics, Mahatma Gandhi University, Kottayam (PIN: 686560), Kerala, India, 2025-10-29/2026-05-31
3. (offline) MSc dissertation titled *"Some studies on cooperative tasks in multi-agent systems"*, Department of Mathematics, Smt. Indira Gandhi Government First Grade Women's College Sagara, Sagara (PIN: 577401), Karnataka, India, 2025-04-02/2025-08-18
4. (offline) MSc dissertation titled *"Studies on consensus, controllability, and observability in multi-agent systems"*, Department of Mathematics, Smt. Indira Gandhi Government First Grade Women's College Sagara, Sagara (PIN: 577401), Karnataka, India, 2024-06-01/2024-11-07

Conference Presentations

1. (offline) *"Controllability of linear impulsive systems—A graph theory perspective"*, National Conference of Differential Equations and Dynamical Systems, 2019-04-05/2019-04-06, Department of Mathematics, National Institute of Technology Puducherry, Karaikal (PIN: 609609), UT of Puducherry, India
2. (offline) *"Controllability of complex networks involving impulses and time-delay controls"*, Second National Conference on Control and Inverse Problems, 2019-03-01/2019-03-02, Department of Mathematics, Central University of Tamil Nadu, Thiruvarur (PIN: 610005), Tamil Nadu, India
3. (offline) *"A PBH-rank condition for the controllability of linear impulsive systems"*, International Conference on Recent Advances in Pure and Applied Mathematics, 2018-10-23/2018-10-25, Department of Applied Mathematics, Delhi Technological University, New Delhi (PIN: 110042), NCT of Delhi, India
4. (offline) *"Controllability of networked higher dimensional systems with one-dimensional communications having multiple constant time-delays in control"*, International Conference on Advances in Pure & Applied Mathematics @ Madurai Kamaraj University, 2018-09-06/2018-09-08, jointly organized by Madurai Kamaraj University, Madurai (PIN: 625021), Tamil Nadu, India and International Multidisciplinary Research Foundation, Vijayawada (PIN: 521456), Andhra Pradesh, India
5. (offline) *"Controllability of fractional order semilinear systems with multiple time-delays in control"*, International Conference on Algebra and Applied Analysis, 2018-08-09/2018-08-11, Department of Mathematics, Integral University, Lucknow (PIN: 226026), Uttar Pradesh, India

Conference Participation

1. (online) "International Conference on Higher Education, Research & Innovation", 2022-07-28/2022-07-30, S & T Digital, Indian Institute of Science Education and Research Pune, Pune (PIN: 411008), Maharashtra, India
2. (offline) "International Conference on Mathematical Analysis and its Applications", 2016-11-28/2016-12-02, Department of Mathematics, Indian Institute of Technology Roorkee, Roorkee (PIN: 247667), Uttarakhand, India
3. (offline) "International Conference on Nonlinear Dynamical Systems" @ Bharathiar University, 2016-03-24/2016-03-26, jointly organized by Bharathiar University and Anna University Regional Campus, Coimbatore (PIN: 641046), Tamil Nadu, India

Review Service

1. Reviewer for ISA Transactions (publication of Elsevier B.V.)
2. Reviewer for IET Control Theory & Applications (publication of John Wiley & Sons)
3. Reviewer for Mathematical Reviews–AMS (reviewer no.: 156160)

Professional Service

1. Convener, "An Hour with Veena V. S.", 2026-10-01, Department of Sciences, Smt. Indira Gandhi Government First Grade Women's College Sagara, Sagara (PIN: 577401), Karnataka, India. URL: https://youtu.be/Vl97XIvQKTI
2. Convener, "An Hour with K. G. Arun", 2026-09-18, Department of Sciences, Smt. Indira Gandhi Government First Grade Women's College Sagara, Sagara (PIN: 577401), Karnataka, India. URL: https://youtu.be/DeSmYcbFxPw
3. Moderator, "Online Talk by Abhipsa Pal", 2026-09-17, Department of Economics, Commerce & Management, Smt. Indira Gandhi Government First Grade Women's College Sagara, Sagara (PIN: 577401), Karnataka, India
4. Moderator, "Online Talk by Pritha Dev", 2026-09-16, Department of Economics, Commerce & Management, Smt. Indira Gandhi Government First Grade Women's College Sagara, Sagara (PIN: 577401), Karnataka, India
5. Convener cum Moderator, "An Hour with Enrique Zuazua", 2026-08-06, Department of Sciences, Smt. Indira Gandhi Government First Grade Women's College Sagara, Sagara (PIN: 577401), Karnataka, India. URL: https://youtu.be/eH5pvsaVoas
6. Technical coordinator for webinars, Smt. Indira Gandhi Government First Grade Women's College Sagara, Sagara (PIN: 577401), Karnataka, India
   - (national) "Stock Market Volatility and Mutual Fund Investment Opportunities", 2025-10-13, Department of Commerce
   - (national) "Trump's Tariffs, GST 2.0, and Fiscal Reforms in India", 2025-10-10, Department of Economics

- (state) “Methodology and Methods of Social Research”, 2025-09-26, Research Cell
- (state) “Multiculturalism”, 2025-09-25, Department of Political Science
- (state) “Political Theory and Politics”, 2025-09-24, Department of Political Science

7. Convener cum Moderator, “Webinar on Cosmos—Into the Unknown”, 2025-08-20/2025-08-22, Department of Sciences, Smt. Indira Gandhi Government First Grade Women’s College Sagara, Sagara (PIN: 577401), Karnataka, India. URLs: https://youtu.be/0gbRvIYGJj8; https://youtu.be/m8CvJq_v-wE

8. Member, “Board of Examinations for 2025–26 Undergraduate Mathematics and Statistics (Regular & Distance)”, 2025-07-03, Kuvempu University, Shivamogga (PIN: 577451), Karnataka, India

9. Question paper setter, “MSc Mathematics – II Semester Examinations – Oct 2024”, 2024-09-26, Sharnbasva University, Kalaburagi (PIN: 585103), Karnataka, India

10. Question paper setter, “MSc Mathematics – I Semester Examinations – Jun 2024”, 2024-04-16, Sharnbasva University, Kalaburagi (PIN: 585103), Karnataka, India

11. Coordinator, “Webinar on Strategies to Crack Competitive Examination after Graduation & Postgraduation @ Sri H. D. Devegowda Government First Grade College Paduvalahippe”, 2023-07-18, IFAS, Pune (PIN: 411008), Maharashtra, India

12. (offline) Convener, “Mathemattitude 2023—A State Level Teaching Competition”, 2023-06-12, Department of Mathematics, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India

13. (offline) Chief guest, “Annual Special Camp @ Teranya”, 2022-06-10, jointly organized by Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe and Model Residential Government First Grade College for Women Haradanahalli, Haradanahalli, (PIN: 573211), Karnataka, India

14. Organizing member, “State Level Webinar on Writing Skills: Commercial Writing”, 2023-02-15, Department of English, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India

15. Faculty coordinator, “World is Full of Networks! An International Webinar on the Applications of Graph Theory”, 2023-01-27, Department of Mathematics, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India

16. MSc Mathematics – I semester admission work, 2023-01-11, University of Mysore, Mysuru (PIN: 570006), Karnataka, India

17. MSc Mathematics – Evaluation work, 2022-09-12/2022-09-13, 2022-09-19/2022-09-20, University of Mysore, Mysuru (PIN: 570006), Karnataka, India

18. MSc Mathematics – Evaluation work, 2022-08-01/2022-08-02, 2022-08-06, 2022-08-16, University of Mysore, Mysuru (PIN: 570006), Karnataka, India

19. Question paper setter for MSc Mathematics, “IV Semester Examinations – Aug 2022 and II Semester Examinations – Oct 2022”, 2022-06-16, University of Mysore, Mysuru (PIN: 570006), Karnataka, India

20. MSc Mathematics – Evaluation work, 2022-04-11/2022-04-15, University of Mysore, Mysuru (PIN: 570006), Karnataka, India

21. (offline) Chief guest, "Annual Special Camp @ Parasanahalli", 2022-03-04, jointly organized by Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe and Model Residential Government First Grade College for Women Haradanahalli, Haradanahalli, (PIN: 573211), Karnataka, India

22. MSc Mathematics – I semester admission work, 2022-02-15, University of Mysore, Mysuru (PIN: 570006), Karnataka, India

23. MSc Mathematics – I semester admission work, 2022-01-05, 2022-01-07, 2022-01-12, 2022-01-18, 2022-01-20, 2022-01-24, University of Mysore, Mysuru (PIN: 570006), Karnataka, India

24. Organizing committee member, "National Webinar on Mathematics and Ramanujan", 2021-12-27, jointly organized by Sarada Vilas College, Mysuru (PIN: 570004) and Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India

25. MSc Mathematics – Evaluation work, 2021-11-09/2021-11-13, University of Mysore, Mysuru (PIN: 570006), Karnataka, India

26. Organizing committee member for webinars, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India

    - (national) "Enriching Lives with Soft Skills", 2021-08-30, Department of English
    - (national) "Role of Social Media in the Political Socialization of Youth in India", 2021-08-16, Department of Political Science
    - (national) "Recent Trends in Materials Research", 2021-08-06, Department of Physics and Chemistry
    - (national) "Challenges of Teaching Kannada Language", 2021-07-17
    - (national) "Challenges of Entrepreneurship", 2021-07-10
    - (national) "Awareness of Intellectual Property Rights", 2021-06-28/2021-07-01

27. (online) Organizing committee member, "National Workshop on 7th International Day of Yoga", 2021-06-21, jointly organized by Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe and Model Residential Government First Grade College for Women Haradanahalli, Haradanahalli, (PIN: 573211), Karnataka, India

28. MSc Mathematics – I semester admission work, 2021-01-13, University of Mysore, Mysuru (PIN: 570006), Karnataka, India

29. Internal member, "Board of Examiners in Mathematics—Postgraduate Degree Examinations, May/Jun 2020 and Dec 2020", 2020-03-02, University of Mysore, Mysuru (PIN: 570006), Karnataka, India

30. (offline) Coordinator, "Mathemattitude 2020—State Level Mathematics Teaching Competition", 2020-02-14, Department of PG Studies in Mathematics, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India

31. (offline) Chief guest, "Social Work Camp @ Madalapura", 2020-02-08, Department of PG Studies in Social Work, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India

32. MSc Mathematics – I semester admission work, 2019-08-07, 2019-08-14, 2019-08-19, University of Mysore, Mysuru (PIN: 570006), Karnataka, India
33. (offline) Trainer, "Life Skill Awareness", 2019-02-20, Department of English, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India
34. (offline) Organizing secretary, "Mathemattitude—An Inter-Collegiate Mathematics Teaching Competition", 2019-02-13, Department of PG Studies in Mathematics, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India
35. (offline) Organizing committee member, "One Day National Level Workshop on Goods and Services Tax", 2018-11-16, Department of PG Studies in Commerce, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India
36. (offline) Trainer, "Life Skill Training Programme", 2018-04-12, Department of Mathematics, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India
37. (offline) Organizing secretary, "Mathemattitude—An Inter-Collegiate Mathematics Teaching Competition", 2018-03-22, Department of PG Studies in Mathematics, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India
38. (offline) Organizing committee member, "One Day State Level Workshop on Legal System in India", 2017-11-21, Department of PG Studies in Social Work, Sri H. D. Devegowda Government First Grade College Paduvalahippe, Paduvalahippe (PIN: 573211), Karnataka, India
39. (offline) Local organizing committee member, "Three-Day National Workshop on Basic MATLAB Programming and Simulink", 2015-05-06/2015-05-08, Department of Mathematics, Central University of Karnataka, Kadaganchi (PIN: 585367), Karnataka, India

Professional Memberships

1. Life-time membership of ISIAM (membership no.: M-65)
2. Outreach membership of SIAM (membership no.: 020833459): 2019-01-01/2019-12-31
3. IEEE membership and young professionals (membership no.: 95418678): 2019-01-01/2019-12-31

Professional Development

1. (online) Participant, "UGC-MMTTC's Two-Week Refresher Course in Mathematics and Its Applications", 2025-12-29/2026-01-12, Department of Mathematics, Assam University, Silchar (PIN: 788011), Assam, India

2. (offline) Participant, "Awareness Workshop on Sarswati Scholarship Scheme @ B. M. S. College of Engineering, Bengaluru", 2025-03-08, AICTE, New Delhi (PIN: 110070), NCT of Delhi, India

3. (offline) Participant, "UGC-MMTTC's Refresher Programme in Mathematics & Computational Sciences", 2025-02-15/2025-02-28, Department of Mathematics, Savitribai Phule Pune University, Pune (PIN: 411007), Maharashtra, India

4. (offline) Participant, "7 Days Vruthi Chaitanya Training Program on Teaching & Learning", 2024-12-16/2024-12-22, Karnataka State Higher Education Academy, Dharwad (PIN: 580003), Karnataka, India

5. (online) Participant, "UGC-MMTTC's Two-Week Interdisciplinary Refresher Course on Recent Advancements in Mathematics, Statistics and Computer Science", 2024-10-14/2024-10-26, Central University of South Bihar, Gaya (PIN: 824236), Bihar, India

6. (offline) Participant, "Two-Week GIAN Course on Network Sciences and Multi-Agent Systems", 2023-06-19/2023-06-30, Department of Mathematical Sciences, Indian Institute of Technology (BHU), Varanasi (PIN: 221005), Uttar Pradesh, India

7. (online) Participant, "Two-Week Refresher Course in Mathematical Sciences", 2021-06-14/2021-06-28, Teaching Learning Centre, Ramanujan College (constituent of University of Delhi), New Delhi (PIN: 110019), NCT of Delhi, India

8. (online & offline) Participant, "Induction Training Programme-45", 2020-12-08/2020-12-18, 2021-03-22/2021-04-01, Karnataka State Higher Education Academy, Dharwad (PIN: 580003), Karnataka, India

9. (online) Participant, "One Week International Faculty Development Programme on Fixed Point Theory and its Applications", 2020-09-15/2020-09-19, Department of Mathematics and Statistics, Manipal University Jaipur, Jaipur (PIN: 303007), Rajasthan, India

10. Participant in MOOCs on,

    - "Differential Equations Part I Basic Theory", 2020-03-25/2020-05-27, Korea Advanced Institute of Science and Technology, Daejeon (postal code: 34141), Republic of Korea
    - "Introduction to Calculus", 2020-04-20/2020-05-25, University of Sydney, Sydney (postcode: 2006), Australia
    - "Mathematics for Machine Learning: Linear Algebra", 2020-04-15/2020-05-20, Imperial College London, London (postcode: SW7 2AZ), United Kingdom
    - "Mathematics for Machine Learning: Multivariate Calculus", 2020-04-07/2020-05-19, Imperial College London, London (postcode: SW7 2AZ), United Kingdom
    - "Differential Equations for Engineers", 2020-04-02/2020-05-14, The Hong Kong University of Science and Technology, Kowloon (postal code: 999077), Hong Kong
    - "Matrix Algebra for Engineers", 2020-04-11/2020-05-09, The Hong Kong University of Science and Technology, Kowloon (postal code: 999077), Hong Kong

11. Participant, "Webinar on How to Write a Quality Technical Paper", 2020-05-10, ISRO Libraries, India

12. (offline) Participant, "Seva Sindhu and AISHE: Start-up Training", 2020-02-29, College Development Council, University of Mysore, Mysuru (PIN: 570006), Karnataka, India

13. (offline) Participant, “Orientation Programme on Implementation of Online Scholarship Portal”, 2019-09-18, College Development Council, University of Mysore, Mysuru (PIN: 570006), Karnataka, India
14. Qualified “Computer Literacy Test for the Employees of Government of Karnataka”, 2018-09-02
15. Qualified “Karnataka Public Service Commission’s Departmental Examination”, 2018-03-28
16. (offline) Participant, “41st Orientation Programme”, 2018-01-22/2018-02-19, UGC-HRDC, Sant Gadge Baba Amravati University, Amravati (PIN: 444602), Maharashtra, India

Social Service

1. Donation of ₹500 towards Kokan Kala & Shikshan Vikas Sanstha, 2024-10-25
2. Donation of ₹500 towards Kokan Kala Va Shikshan Vikas Sanstha, 2024-06-29 (receipt no. KKN-144721)
3. Donation of ₹500 towards Khushii Kinship for Humanitarian Social and Holistic Intervention, 2023-06-23 (receipt no. KHUSHII/8503/14304)
4. Donation of ₹500 towards Kailash Satyarthi Children’s Foundation, 2022-03-17 (receipt no. 3483)
5. Donation of ₹500 towards World Wildlife Fund-India, 2022-03-11 (receipt no. 026234)
6. Donation of ₹1,000 towards World Wildlife Fund-India, 2021-09-17 (receipt no. 012042)
7. Donation of ₹500 towards PM CARES Fund during COVID-19, 2020-04-04 (order no. 10677409750)
8. Donation of ₹1,952 towards Karnataka CM Relief Fund, 2019-08
9. Donation of ₹1,869 towards Karnataka CM Relief Fund, 2018-09